

StorePulse Tasks

1. Define project scope

Agree on the exact topic, main goal and what the project will and will not include. For example, the system will analyze retail sales data, but it will not include user authentication, live updates or machine learning.

2. Identify business questions

Decide which questions the analytics should answer. For example: What are the top-selling products? How do sales change over time? Which categories perform best? What is the average order value?

3. Design the data model

Define the main entities and their relationships: customers, products, orders and order items. This step ensures the structure is logical before implementation starts.

4. Create database schema

Write SQL script(s) to create tables with appropriate columns, primary keys and foreign keys. Add simple constraints where needed to keep data consistent.

5. Plan synthetic data rules

Decide how the fake data should look so it feels realistic. For example, products should have categories and prices, orders should belong to customers and order items should connect products with quantities.

6. Generate synthetic customers

Create a dataset of fake customers with fields like name, city and registration date. The goal is to have enough variety for meaningful analysis.

7. Generate synthetic products

Create a list of products with names, categories and prices. Make sure the product catalog is small enough to manage but large enough to produce useful charts.

8. Generate synthetic orders

Create order records with customer links and order dates. These records should simulate purchases over a period of time so monthly or weekly analysis is possible.

9. Generate order items

For each order, create one or more order item rows with product, quantity and item price. This table is important because it makes revenue analysis more realistic.

10. Export or save raw data

Store the generated data in CSV files or directly prepare it for database import. This makes the pipeline easier to reproduce and test.

11. Load data into the database

Import the synthetic data into SQLite or PostgreSQL. After loading, check whether all tables contain the expected number of records.

12. Check data integrity

Verify that relations between tables are correct. For example, every order should belong to a valid customer and every order item should reference an existing order and product.

13. Write basic SQL queries

Prepare queries for core metrics such as total revenue, number of orders, average order value, monthly sales and top categories.

14. Write advanced SQL queries

Add slightly more detailed queries such as top customers, top products by revenue, sales by city, or category comparison. These make the project look more complete.

15. Test SQL outputs

Make sure the queries return correct and reasonable results. This helps catch problems in the data model or data generation logic.

16. Load query results into Python

Use pandas to read the SQL results or database tables into Python for further analysis.

17. Clean and prepare data in Python

Remove duplicates if needed, convert date columns, calculate derived metrics and make the dataset ready for visualization.

18. Perform exploratory analysis

Look at patterns in the data: sales trends, category performance, customer activity and distribution of order values.

19. Create visualizations

Build charts such as line charts, bar charts, histograms and maybe a pie chart. The charts should clearly show the main findings.

20. Refine chart readability

Add titles, labels, legends and proper formatting so the charts are easy to understand without extra explanation.

21. Extract key insights

Write a short summary of what the analysis shows. For example, which products generate the most revenue, whether sales are seasonal and which customer groups are most active.

22. Prepare final report

Put together a short report with the project description, data model, methods, charts and conclusions.

23. Write README instructions

Explain how to run the project step by step: install dependencies, generate data, create database tables, run analysis and view charts.

24. Final code cleanup

Organize files, rename scripts clearly, remove unused code and make sure everything is easy to understand.

25. End-to-end testing

Run the whole pipeline from start to finish to confirm that data generation, database loading, SQL analysis and charts all work together.